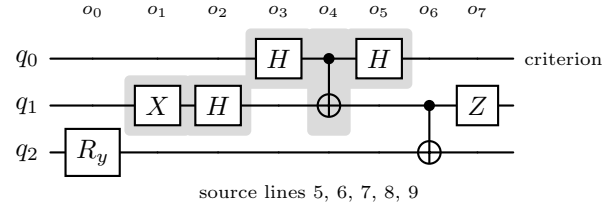
(a) Qiskit program

```
1  from qiskit import QuantumCircuit
2
3  qc = QuantumCircuit(3)
4  qc.ry(0.9, 2)        # o0
5  qc.x(1)              # o1
6  qc.h(1)              # o2
7  qc.h(0)              # o3
8  qc.cx(0, 1)          # o4  kickback onto q0
9  qc.h(0)              # o5
10 qc.cx(1, 2)          # o6  q1, q2 entangled
11 qc.z(1)              # o7
```

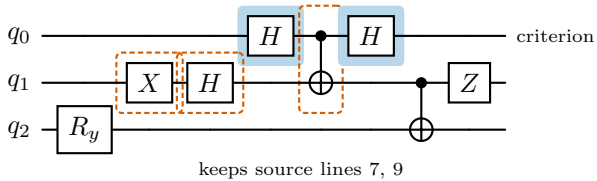
(b) Dependences of q_0 (ground truth)



source lines 5, 6, 7, 8, 9

operations where a change can alter q_0's state

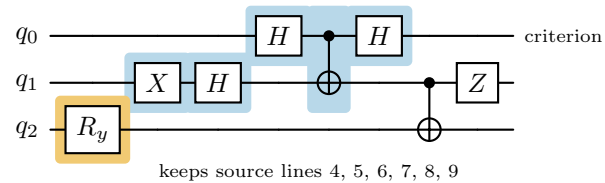
(c) Def-use slice of q_0



keeps source lines 7, 9

fidelity 0.00 at $|000\rangle$; wrong at 20 of 20 random inputs

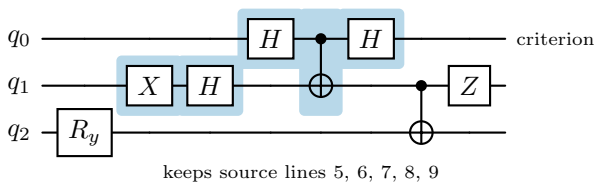
(d) Entanglement-closure (QPDG) slice of q_0



keeps source lines 4, 5, 6, 7, 8, 9

fidelity 1.00 at $|000\rangle$; correct at all 20 random inputs

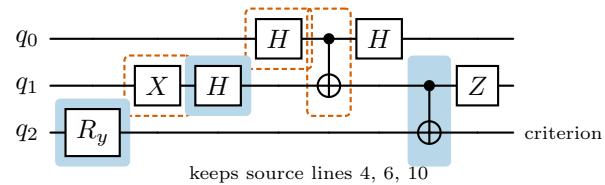
(e) Wire-rule slice of q_0



keeps source lines 5, 6, 7, 8, 9

fidelity 1.00 at $|000\rangle$; correct at all 20 random inputs

(f) State-based (QSlice) slice of q_2 at $|000\rangle$



keeps source lines 4, 6, 10

fidelity 1.00 at $|000\rangle$; wrong at 20 of 20 random inputs

dependence | in slice and a dependence | in slice, not a dependence | missed dependence